METRIC
MIL-DTL-155C
w/AMENDMENT 2
06 February 2025
SUPERSEDING
MIL-DTL-155C
w/ AMENDMENT 1
21 November 2023

DETAIL SPECIFICATION

GRAPHITE, DRY (For use in ammunition)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations and tests for commercial grade graphite and is intended for use in ammunition.

1.2 Classification. The graphite should be of the following grades:

Grade I	-	Manufactured graphite used for lubrication
Grade II	-	Natural graphite used for lubrication
Grade III	-	Manufactured graphite used for glazing
Grade IV	-	Natural graphite used for glazing

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to: U.S. Army Armaments Center, ATTN: FCDD-ACE-QSA, Picatinny Arsenal, New Jersey 07806-5000, or emailed to usarmy.pica.devcom-ac.list.ardec-stdzn-branch@army.mil. Since contact information can change, you may want to verify the currency of this information using ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 1376

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:19Z
Check the source to verify that this is the current version before use.

2.2 Government documents

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916	-	DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at https://quicksearch.dla.mil/).

2.3 Non-Government standards and publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM E11	-	Standard Specification for Woven Wire Test Sieve Cloth and Test Sieves
ASTM E300	-	Standard Practice for Sampling Industrial Chemicals

(Copies of these documents are available from https://www.astm.org/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Required inspections.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.2 Conformance. A sample shall be subject to conformance inspection in accordance with 4.3.

3.2 Appearance (applicable to Grades II and IV only). Graphite of Grades II and IV shall appear steel gray or silver gray in color and shall have a metallic luster.

3.3 Chemical and physical properties. Graphite shall conform to the physical and chemical requirements specified in Table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:19Z
Check the source to verify that this is the current version before use.

TABLE I. Chemical and physical requirements for graphite.

Properties	Grade I percent	Grade II percent	Grade III percent	Grade IV percent
Moisture, maximum (max.)	0.20	0.50	0.50	0.50
Ash, max.	0.60	6.0	0.60	6.0
Silica, max.	-	1.0	-	2.75
Grit	None	None	None	None
Acidity	None	None	None	None
Free sulfur, max.	0.02	-	0.05	0.05
Total sulfur, max.	0.20	-	0.50	0.50
Granulation				
Through U.S. Alternative No. 100 (149 micron), minimum (min.)	-	65.0	-	-
Through U.S. Alternative No. 200 (74 micron), min.	96.0	-	96.0	-
Through U.S. Alternative No. 325 (44 micron), min.	-	-	-	96.0

3.4 Lubricating quality (applicable to Grades I and II only). When specified (see 6.2 and 6.7), Grades I and II graphite shall be suitable for use as a pelletizing lubricant. The graphite shall be considered satisfactory if the machine works smoothly and easily, if the powder does not stick to any parts of the machine, and if the resulting pellets do not fall apart. Additional requirements may be specified in the contract.

3.5 Glazing quality (applicable to Grades III and IV only). When specified (see 6.2 and 6.7), Grades III and IV graphite shall be suitable for use as a glazing compound. The graphite shall be considered satisfactory if it is processed through the glazing equipment and provides a uniform coating. Additional requirements may be specified in the contract, such as surface texture and magnification.

3.6 Workmanship. Graphite shall be free from visible impurities and foreign material such as dirt, rust, paint, wooden particles or any other foreign objects.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:19Z
Check the source to verify that this is the current version before use.

4. VERIFICATION

TABLE II. Requirements/verification cross reference matrix.

Method of verification 1 – Analysis 2 – Demonstration 3 – Examination 4 – Test						Classes of verification A – First article B – Conformance		
Section 3 Requirement	Description	Verification Methods				Verification Class		Section 4 Verification
		1	2	3	4	A	B	
3.1.1	First article		X	X	X	X		4.2
3.1.2	Conformance			X	X		X	4.3
3.2	Appearance			X		X	X	4.4.11
3.3	Moisture				X	X	X	4.4.1
3.3	Ash				X	X	X	4.4.2
3.3	Silica				X	X	X	4.4.3
3.3	Grit				X	X	X	4.4.4
3.3	Acidity				X	X	X	4.4.5
3.3	Free sulfur				X	X	X	4.4.6
3.3	Total sulfur				X	X	X	4.4.7
3.3	Granulation				X	X	X	4.4.8
3.4	Lubricating quality		X			X		4.4.9
3.5	Glazing quality		X			X		4.4.10
3.6	Workmanship			X		X	X	4.4.11

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see paragraph 4.2)
b. Conformance inspection (see paragraph 4.3)

4.2 First article inspection. When specified (see 6.2), a sample shall be subjected to first article verification in accordance with Table III. Samples of not less than 7 kilograms (15 pounds) of graphite collected in accordance with 4.3.2 shall be submitted for examination. Samples shall be produced by the contractor using the same process procedures, material supply sources, production equipment, and production line as will be used in regular production.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:19Z
Check the source to verify that this is the current version before use.

TABLE III. First article acceptance tests.

Category	Examination or test	Requirement paragraph	Inspection method reference
Critical	None defined		
Major			
101	Appearance	3.2	4.4.11
102	Moisture	3.3	4.4.1
103	Ash	3.3	4.4.2
104	Silica	3.3	4.4.3
105	Grit	3.3	4.4.4
106	Acidity	3.3	4.4.5
107	Free sulfur	3.3	4.4.6
108	Total sulfur	3.3	4.4.7
109	Granulation	3.3	4.4.8
110	Lubricating quality	3.4	4.4.9
111	Glazing quality	3.5	4.4.10
Minor			
201	Workmanship	3.6	4.4.11

4.2.1 Rejection. If any sample fails to comply with any of the applicable requirements the first article quantity shall be rejected.

4.3 Conformance inspection.

4.3.1 Inspection lot formation. Inspection lots and batches shall comply with the MIL-STD-1916, paragraph "Formation and identification of lots or batches." Each batch shall consist of a quantity of graphite that has been subjected to the same chemical and physical mixing process intended to make the final product homogeneous. A lot shall consist of one acceptable batch (see 6.3.1) of graphite produced by one manufacturer in accordance with one specification and revision under one continuous set of operating conditions.

4.3.2 Graphite sampling. Sampling for inspection shall be performed at either the supplier/distributor location where the graphite is packed, or at the manufacturer's receiving facility where the material is unpacked for use in the next higher sub-assembly or formulation.

4.3.2.1 Sampling at the supplier/distributor. Samples shall be collected from the beginning, middle, and end of every batch in every lot in accordance with ASTM E300. A composite sample of not less than 225 grams (g) shall be formed which is assumed to be representative of that batch. All composite samples shall be verified in accordance with 4.4. If any sample fails to comply with any applicable requirements the batch shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:19Z
Check the source to verify that this is the current version before use.

4.3.2.2 Sampling at the receiving facility. Graphite packaged in shipping containers (e.g., drums, boxes, etc.) shall be sampled using Verification Level II of MIL-STD-1916. Each shipping container shall be sampled in three locations in accordance with ASTM E300 using a thief stick. The samples from each shipping container shall then be placed in a secondary airtight container and mixed to form a composite sample of not less than 225 g for each graphite lot received. There shall be no reduced sampling. Every lot received shall be tested in accordance with 4.4. If any sample fails to meet any test requirement, the lot represented by the sample shall be rejected.

4.3.3 Classification of characteristics. Graphite shall be subjected to the inspections according to 4.3.3.1.

4.3.3.1 Graphite.

Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Appearance	VL II	3.2	4.4.11
102	Moisture	4.3.2	3.3	4.4.1
103	Ash	4.3.2	3.3	4.4.2
104	Silica	4.3.2	3.3	4.4.3
105	Grit	4.3.2	3.3	4.4.4
106	Acidity	4.3.2	3.3	4.4.5
107	Free sulfur	4.3.2	3.3	4.4.6
108	Total sulfur	4.3.2	3.3	4.4.7
109	Granulation	4.3.2	3.3	4.4.8
Minor				
201	Workmanship	VL II	3.6	4.4.11

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:19Z
Check the source to verify that this is the current version before use.

4.4 Test methods and procedures.

4.4.1 Determination of moisture. Transfer a 10.0 ± 0.5 g portion of the composite sample to a weighing dish and record the gross weight to the nearest 0.1 milligram (mg). Dry the dish and contents for 1 hour at 100 – 105 degrees Celsius (°C) until gross weights from two consecutive 1-hour drying cycles differ by no more than 1.0 mg, cool the dish in a desiccator for 1 hour at room temperature (20 – 22 °C) and weigh. Calculate the loss in weight and percent of moisture.

$$\text{Percent moisture} = \frac{(B - A)}{W} * 100$$

Where:

A = weight of dish and sample after heating, g
B = weight of dish and sample before heating, g
W = weight of sample, g

4.4.1.1 Alternate method for determination of moisture. Determine moisture by use of a dedicated moisture analyzer or other standard method as approved by the contracting agency (see 6.5).

4.4.2 Determination of ash. Transfer a 2.00 ± 0.05 g portion of the composite sample to a previously ignited crucible pre-weighed to the nearest 0.1 mg (use a porcelain crucible for Grades I and III, platinum for Grades II and IV). Ignite the crucible and contents in a muffle furnace at 950 ± 50 °C with an open air vent/exhaust or other means for maintaining flow of air across the sample. If the furnace is to remain hot while the crucible is inserted or removed, use proper heat-protective personal protective equipment (PPE) and tongs to transfer the crucible. Combust for approximately 4 hours, until all combustible material is gone. Remove the crucible from the muffle furnace and place it on a heat-proof surface until it is cool enough to place in a desiccator. Cool the crucible to room temperature in the desiccator and reweigh. Calculate the percentage ash as follows:

$$\text{Percent ash} = \frac{(C - D)}{W} * 100$$

Where:

C = weight of crucible and residue after heating, g
D = weight of crucible, g
W = weight of sample, g

NOTE. Retain the ash residue for the silica and other grit determinations as specified in 4.4.3 and 4.4.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:19Z
Check the source to verify that this is the current version before use.

4.4.3 Determination of silica. (Applicable to Grades II and IV only). To the ash (see 4.4.2) in the platinum crucible, add 5 milliliters (mL) of hydrofluoric acid and a few drops of sulfuric acid. Evaporate to dryness atop a hotplate and ignite in a muffle furnace at 950 ± 50 °C for 4 hours. Place the crucible on a heat-proof surface until cool enough to be placed in a desiccator. Cool in a desiccator to room temperature, then reweigh to four decimal places. Calculate the loss in weight as percent silica as follows, and retain the residue for grit determination.

$$\text{Percent silica} = \frac{(C - E)}{D} * 100$$

Where:

C = weight of crucible + residue before acid treatment, g
D = weight of graphite sample used for ash determination, g
E = weight of crucible + residue after ignition, g

4.4.4 Determination of grit.

4.4.4.1 Determination of grit for Grade I and Grade III. Transfer at least three portions of approximately 0.1 g each of the composite sample, to smooth glass slides and cover each with a clean glass slide. Rub each material between glass slides determining the presence of grit by scratching noise and or scratches on the glass slides.

4.4.4.2 Determination of grit for Grade II and Grade IV. Transfer the residue from silica determination to a smooth glass slide and cover with another clean glass slide. Rub the material between the glass slides and determine the presence of grit by scratching noise and or scratches on the glass slides.

4.4.5 Determination of acidity. Transfer a portion of approximately 10 g of the composite sample to a 250 mL beaker, add 100 mL of neutral distilled water and heat quickly to boiling while stirring. Filter immediately and cool to room temperature. Test the filtrate by adding two drops of phenolphthalein and then two drops of methyl red. If colorless to phenolphthalein and yellow to methyl red, acidity shall be reported as none.

4.4.6 Determination of free sulfur.

4.4.6.1 High pressure liquid chromatography (HPLC) method.

4.4.6.1.1 Reagents.

a. Water, HPLC grade
b. Acetonitrile, HPLC grade
c. Tetrahydrofuran (THF), unstabilized, reagent or HPLC grade
d. Sulfur, reagent grade

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:19Z
Check the source to verify that this is the current version before use.

4.4.6.1.2 Supplies and equipment.

a. Volumetric flasks, glass, 1 x 50 mL capacity, 3 x 25 mL capacity Class A or B
b. Volumetric pipettes, glass 1 mL, 2 mL, 3 mL, 10 mL, one (1) each, all Class A or B
c. Test tubes, glass, 20-24 millimeter (mm) inside diameter (ID), 150 - 200 mm long
d. Ultrasonic bath
e. Analytical balance, ≥10 g capacity with 0.0001 g resolution or finer
f. High or ultrahigh pressure liquid chromatograph (HPLC or UPLC) with photodiode array detector with a 10 mm path length cell
g. HPLC column, RP-18, 50 x 2.1 mm with 2.7 – 5 micrometer (µm) particle size, or equivalent
h. Syringe, glass or polypropylene, 3-5 mL capacity, luer tip or luer lock
i. Syringe filter (PTFE), 25-35 mm diameter, 0.1 - 0.45 µm porosity
j. HPLC vials

4.4.6.1.3 Sample preparation. The sample shall be prepared in duplicate. For each of the duplicates, transfer a 2.50 ± 0.10 g portion of the composite sample to a test tube. Add 10 mL of THF using a volumetric pipette. Allow 1-2 minutes for the solvent to infuse, and mark the liquid level with a permanent marker. Position the test tubes in an ultrasonic bath at a 45-60° angle from vertical (e.g. in a beaker partially filled with water). Sonicate the test tubes for 30 minutes without cooling. Check the liquid level; if level has dropped by more than 5 mm refill to mark with THF. Continue sonication for another 30 minutes. Remove test tubes and rest vertically for 30 minutes. Check liquid level and fill up to mark with THF. Swirl test tubes to thoroughly mix and allow to settle for 10 minutes. Decant clear fraction into filtering syringe and filter into HPLC vial.

4.4.6.1.4 Standard preparation.

a. Prepare an approximate 1150 milligram/liter (mg/L) sulfur stock solution by dissolving 55.0 - 60.0 mg of sulfur in THF in a 50 mL volumetric flask. Fill to volume with THF and mix.
b. Prepare three working standards, with appx. concentrations 46 mg/L, 92 mg/L, 138 mg/L. Pipet 1.00 mL, 2.00 mL, and 3.00 mL aliquots of the stock solution into three 25 mL volumetric flasks. Fill each to volume with THF and mix. Calculate the precise sulfur concentrations in mg/L to the nearest 0.1 mg/L. Filter each standard through a filtering syringe into an HPLC vial.

4.4.6.1.5 Preparation of calibration curve.

a. Condition HPLC column at 35°C, running 0.25 mL/minute of an 80:20 mix (by volume) of acetonitrile and water until a steady baseline is obtained. Note: temperature, flowrate, and eluent composition may be modified slightly to allow sulfur to be sufficiently resolved from other components and to elute within a reasonable time (e.g. 10 minutes).
b. Make three (3) 2-µL injections of each standard prepared in 4.4.6.1.4b and one (1) injection of THF.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:19Z
Check the source to verify that this is the current version before use.

c. Collect the UV spectra between 200-400 nm once per second at maximum available resolution.
d. Extract chromatograms at 225 nanometer (nm). Identify the sulfur peak. Under the conditions in "a", sulfur (cyclooctasulfur) elutes at 5.1 ± 0.2 min. Its ultraviolet spectrum shall closely match that given in Figure 1; the spectrum shall contain local minima at 210-215 nm, 245-255 nm, a local maximum at 220-227 nm and a broad local maximum at 255-280 nm. Alternately calculate the area ratio, of the suspected sulfur peak, from chromatograms extracted at 225 nm to same at 261 nm; the ratio shall be within 1.63 ± 0.08.
e. Chromatograms extracted at 225 nm shall be used to construct a linear calibration relating sulfur peak areas to concentrations. Repeat steps a-e if the correlation coefficient (R^2) is <0.999.

4.4.6.1.6 Sample analysis.

a. The duplicate samples prepared in 4.4.6.1.3 shall be analyzed in the same manner as the standards, i.e. using the steps 4.4.6.1.5 a-d, except one injection per sample shall suffice.
b. Sample chromatograms extracted at 225 nm and the calibration from 4.4.6.1.5e shall be used to calculate sulfur concentrations for each injection (i.e. each duplicate sample preparation) based on the peak area of sulfur.
c. Calculate percent free sulfur for each duplicate sample preparation as follows:

$$\text{Percent free sulfur} = \frac{C}{1000 * W}$$

Where:

C = sulfur concentration, mg/L
W = sample weight from 4.4.6.1.3, g

d. Average the duplicate results and report the average as percent free sulfur.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:19Z
Check the source to verify that this is the current version before use.

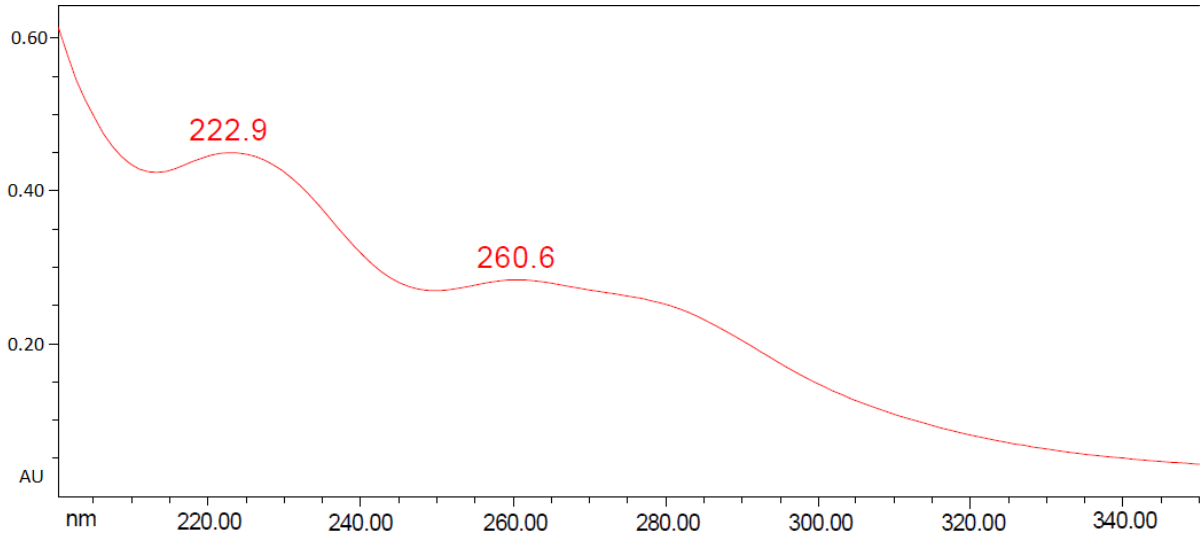


FIGURE 1. Ultraviolet spectrum of S_8 sulfur (see 4.4.6.1.5d).

4.4.6.2 High performance liquid chromatography (HPLC) alternate method.

4.4.6.2.1 Reagents.

a. Water, HPLC grade
b. Acetonitrile, HPLC grade
c. Tetrahydrofuran (THF), unstabilized, reagent or HPLC grade
d. Sulfur, reagent grade

4.4.6.2.2 Supplies and equipment.

a. Volumetric flasks, glass, 1 x 100 mL capacity, 5 x 25 mL capacity Class A or B
b. Volumetric pipettes, glass, 1 mL, 2 mL, 5 mL, 10 mL, and 20 mL
c. Wrist-action shaker capable of producing 375-400 oscillations / minute
d. Centrifuge
e. Centrifuge tubes with screw caps, 40 mL and 15 mL
f. Pipettes, disposable plastic
g. Analytical balance, 250 g capacity with 0.0001 g resolution or finer
h. High Performance Liquid Chromatograph (HPLC) with UV detector capable of measuring at 225nm
i. HPLC column, RP-18, 50 x 21 mm with 2.7-5 µm particle size
j. Syringe, glass or polypropylene, 3-5 mL capacity, luer tip or luer lock
k. Syringe filter, 25-35 mm diameter, 0.1-0.45 µm porosity
l. HPLC vials and caps

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:19Z
Check the source to verify that this is the current version before use.

4.4.6.2.3 Sample preparation. The sample shall be prepared in duplicate. For each of the duplicates, transfer a 2.50 ± 0.10 g portion of the composite sample, accurately weighed to the nearest 0.1 mg, to a 40 mL centrifuge tube. Add 10 mL of THF using a volumetric pipet. Cap, then place the centrifuge tubes onto the wrist-action shaker and shake under routine laboratory conditions for at least 45 minutes. Remove tubes and place in centrifuge for 20 minutes. Pipette 5 mL of the clear liquid fraction into a 15 mL centrifuge tube and centrifuge for 5 minutes. Decant clear fraction into a syringe/filter assembly and filter the liquid into an HPLC vial.

4.4.6.2.4 Standard preparation.

a. Prepare a 250 mg/L sulfur stock solution by dissolving 25 mg of sulfur in THF in a 100 mL volumetric flask. Fill to volume with THF and mix.
b. Prepare five working standards with approximate concentrations of 10 mg/L, 20 mg/L, 50 mg/L, 100 mg/L, and 200 mg/L. Pipet 1.00 mL, 2.00 mL, 5.00 mL, 10 mL, and 20 mL aliquots of the stock solution into five 25 mL volumetric flasks. Fill each to volume with THF and mix. Calculate the precise sulfur concentrations in mg/L to the nearest 0.1 mg/L. Filter each standard through a syringe/filter assembly into an HPLC vial.

4.4.6.2.5 Preparation of calibration curve.

a. Condition the HPLC column at 35°C, running 0.25 mL/min of an 80:20 mix (by volume) of acetonitrile and water until a steady baseline in obtained. Note: temperature, flowrate and eluent composition may be modified to allow sulfur to be sufficiently resolved from other components and to elute within a reasonable time (e.g. 10 minutes).
b. Inject 2 µL of each standard prepared in 4.4.6.2.4b. and monitor the UV detector output at 225 nm. The sulfur peak elutes at approximately 5.1 minutes under the conditions in section a.
c. Determine the peak area of the sulfur and construct a linear calibration relating the sulfur peak area to sulfur concentration.
d. Repeat steps a-c if the correlation coefficient (R squared) is <0.999.

4.4.6.2.6 Sample analysis.

a. The duplicate samples prepared in 4.4.6.2.3 shall be analyzed in the same manner as the standards, using steps 4.4.6.2.5 a-b.
b. Determine the area of the sulfur peak and use the calibration curve from 4.4.6.2.5c to calculate the sulfur concentration for each injection.
c. Calculate the percent free sulfur for each duplicate sample preparation as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:19Z
Check the source to verify that this is the current version before use.

$$\text{Percent free sulfur} = \frac{C}{1000 * W}$$

Where:

C = sulfur concentration, mg/L
W = sample weight from 4.4.6.2.3, g

d. Average the duplicate results and report the average as percent free sulfur.

4.4.7 Determination of total sulfur.

4.4.7.1 Inductively coupled plasma emission spectrophotometer (ICP-ES) method.

4.4.7.1.1 Reagents. It is advisable to use commercially available stock solutions with certified concentrations traceable to primary standards (National Institute of Standards and Technology or international measurement standards). Stock solutions shall be replaced after one year or in accordance with manufacturer recommendations. Alternately, stock solutions may be prepared in-house from high purity sodium sulfate. The maximum recommended shelf life is one year from date of initial preparation.

a. Water, distilled or de-ionized (DI) water with conductivity below 0.1 microSiemens (uS/cm), (all references to water throughout this section shall be interpreted as such)
b. Certified Sulfur stock solution shall contain the equivalent of 1,000 ± 10 mg/L sulfur dissolved in water
c. Anhydrous sodium sulfate ≥ 99.0% (alternative if preparing stock solution in-house)
d. Nitric acid, 67-70 %wt, ACS or Reagent grade
e. Hydrochloric acid, 34-38 %wt, ACS or Reagent grade
f. Sodium carbonate, ACS or Reagent grade

4.4.7.1.2 Supplies and equipment.

a. Volumetric flasks, 100 mL capacity, Class A or B
b. Volumetric pipettes, 1 mL, 2 mL, 5 mL, all Class A or B
c. Benchtop hotplate
d. Steam bath (optional)
e. Analytical balance, ≥10 g capacity with 0.001 g resolution or finer
f. ICP-ES, with radial view, e.g. ThermoFisher iCap 6000 series, or equivalent

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:19Z
Check the source to verify that this is the current version before use.

4.4.7.1.3 Sample preparation. The sample shall be prepared in duplicate along with a reagent blank as follows. Transfer a 1.000 ± 0.100 g portion of the composite sample to a beaker. Add 30 mL of concentrated nitric acid and 0.10 ± 0.02 g of sulfate-free sodium carbonate. Heat the mixture to boiling and add 60 mL of concentrated hydrochloric acid, slowly at first, until violent reaction has subsided. Concentrate the solution to dryness. Moisten the residue with 2-4 mL of concentrated hydrochloric acid, add 30-40 mL of hot distilled water and digest for a few minutes on a steam bath. Filter the mixture and wash the residue with water, collecting the filtrate in a 100 mL volumetric flask. Make the final volume up to 100 mL with water. Prepare a blank solution using the same procedure but omitting the sample.

4.4.7.1.4 Standard preparation.

a. Prepare a 1000 mg/L sulfur stock solution by dissolving 443 ± 4 mg of anhydrous sodium sulfate in water in a 100 mL volumetric flask. Make up to volume with water. Alternately, use an aqueous, commercial reference solution containing the equivalent of 1000 ± 10 mg/L sulfur.
b. Make a 10 mg/L standard solution. Pipet a 1.00 mL aliquot of the stock solution into a 100 mL volumetric flask. Dilute to volume with water.
c. Make a 20 mg/L standard solution. Pipet a 2.00 mL aliquot of the stock solution into a 100 mL volumetric flask. Dilute to volume with water.
d. Make a 50 mg/L standard solution. Pipet a 5.00 mL aliquot of the stock solution into a 100 mL volumetric flask. Dilute to volume with water.

4.4.7.1.5 Preparation of calibration curve.

a. Sulfur emission intensity shall be measured using radial view at 182.0 nm.
b. Aspirate water, and the 10, 20 and 50 mg/L standard solutions into the ICP spectrometer in order of increasing concentration.
c. Take emission measurements and generate a calibration curve utilizing the instrument's software. The calibration shall be forced through the origin. Repeat the calibration if the correlation coefficient (R^2) is <0.999.

4.4.7.1.6 Sample analysis.

a. Aspirate first replicate sample solution (4.4.7.1.3) into ICP spectrometer and record sulfur emission measurements.
b. Software generated concentrations for sulfur shall be inserted into the formula below as A_1.
c. Aspirate reagent blank solution (4.4.7.1.3) and insert the resulting sulfur concentration into formula below as A_2.
d. Repeat steps a and b for the second replicate sample solution (4.4.7.1.3).
e. Perform the sample calculation, below, for each of the two duplicates thus obtaining two "X" values. Average the two "X" values and report as total sulfur, % by weight.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:19Z
Check the source to verify that this is the current version before use.

$$X = 0.01\frac{(A_1 - A_2)}{W}$$

Where:

X	=	Total sulfur, % by weight
A_1	=	concentration of sulfur detected in sample, mg/L
A_2	=	concentration of sulfur detected in reagent blank solution, mg/L
W	=	sample weight, g

4.4.7.2 Alternate method for determination of total sulfur. Transfer a 1.000 ± 0.100 g portion the composite sample to a beaker. Add 30 mL of concentrated nitric acid and 0.1 g of sulfate-free sodium carbonate. Heat the mixture to boiling and then add 60 mL of concentrated hydrochloric acid, slowly at first, until violent reaction has subsided. Concentrate the solution to dryness. Moisten the residue with 2 to 4 mL of concentrated hydrochloric acid, add 30 to 40 mL of hot distilled water and digest for a few minutes on a steam bath. Filter the mixture and wash the residue with distilled water, keeping the final volume of filtrate washing within 100 mL. Heat the filtrate to boiling and rapidly add 10 mL of 5 percent barium chloride solution, with vigorous agitation. Allow the precipitate of barium sulfate to settle for at least 24 hours. Filter the mixture through a tared fine porosity crucible. Wash the residue with hot distilled water and test the last few washings of the precipitate with 2 drops of silver nitrate to see that all chloride is removed. Heat over a Bunsen burner until dull red-hot, then cool the crucible for 1 hour at room temperature (20 – 22 °C) and weigh. Calculate the increase in weight, corrected for the barium sulfate obtained in a blank determination of all reagents, to percentage of total sulfur in the specimen as follows:

$$\text{Percent total sulfur} = \frac{(H - I)}{Y} * 13.73$$

Where:

H	=	weight of precipitate in sample, g
I	=	weight of precipitate in blank, g
Y	=	weight of sample, g

4.4.8 Determination of granulation. Place a 25.00 ± 0.50 g portion of the composite graphite sample and two metallic washers weighing 2 – 3 g each on the appropriate U.S. standard sieve, meeting the requirements of ASTM E11, assembled to a bottom pan. Place a cover on the sieve and shake the assembly for 10 minutes by hand or 5 minutes by the means of a mechanical shaker geared to produce 292 ± 24 gyrations per minute and 150 ± 10 taps of the striker per minute. Weigh the portion retained on the sieve and calculate the percentage of specimen passing through the sieve as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:19Z
Check the source to verify that this is the current version before use.

$$\text{Percent through} = \frac{(Z - J)}{Z} * 100$$

Where:

J = weight retained on designated sieve, g
Z = weight of sample, g

4.4.9 Lubricating quality. When specified, a sufficient amount of material shall be used to verify the requirement of paragraph 3.4 (see 6.2 and 6.8). If multiple batches of material are to be used in production a sample from each shall be used.

4.4.10 Glazing quality. When specified, a sufficient amount of material shall be used to verify the requirement of paragraph 3.5 (see 6.2 and 6.8). If multiple batches of material are to be used in production a sample from each shall be used.

4.4.11 Workmanship and appearance. Take approximately 10 g of graphite and spread the composition out on a well-lit, clean, white surface. The sample shall be examined with unaided eyes by personnel with normal vision or normal corrected vision. The sample shall be uniform in quality and free from lumps, visible impurities, foreign matter and Grades II and IV shall meet requirements for appearance.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of the materiel is to be performed by DOD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Graphite is intended for the following uses:

a. Grade I and II as lubricants in pelleting explosives.
b. Grade III and IV as glazing agents.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification
b. Issue of ASSIST to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and 2.2)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:19Z
Check the source to verify that this is the current version before use.

c. Provisions for submission/resubmission of first article samples
d. Quantity required and delivery schedules
e. Levels of preservation and packing (see 5.1)
f. Item hazard classification
g. Certificate(s) of Analysis reporting the test results for the batch(es) or container(s) tested
h. Grade of material required (see 1.2)
i. Requirements and tests for lubricating quality, if required, for Grade I and Grade II only (see 3.4, 4.4.9 and 6.8)
j. Requirements and tests for glazing quality, if required, for Grade III and Grade IV only (see 3.5, 4.4.10 and 6.8)

6.3 Definitions.

6.3.1 Batch. A discrete quantity of material that has been manufactured through various unit operations to produce a substantially uniform final product.

6.4 Technical agency. Picatinny Arsenal is the technical agency that is responsible for the preparation and revision of the specification. All correspondence to the Technical Agency should be submitted through the contracting officer to: CCDC AC, ATTN: FCDD-ACE-QEE, Picatinny Arsenal, NJ 07806-5000.

6.5 Approval of equivalent test methods and/or instruments. Prior approval of the Contracting Officer should be required for use of equivalent test methods, instruments and/or reagents. A description of the proposed method, instrumentation and/or reagents should be submitted through the Contracting Officer to the Technical Agency (see 6.4): CCDC AC, ATTN: FCDD-ACE-QEE, Picatinny Arsenal, NJ 07806-5000. This description should include but not be limited to the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

6.6 Submission of alternative conformance inspection provisions. Unless otherwise specified in the contract, proposed alternative conformance inspection provisions should be submitted by the contractor for evaluation by the Technical Agency (see 6.4) responsible for preparation of this specification.

6.7 Historic pelletizing and glazing quality tests. Prior revisions of MIL-G-155 called out methods for performing testing. Specifically desired requirements and tests should be called out on the drawing, contract, or purchase order (see 6.2).

6.7.1 Historic pelletizing quality. Prior revisions specified “Mix two parts by weight of the sample and 100 parts of weight of tetryl complying with the requirements of specification JAN-T-339. Make at least 25 pellets of the mixture by subjecting portions of the mixture to pressure in a pelletizing machine.”

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:19Z
Check the source to verify that this is the current version before use.

6.7.2 Historic glazing quality. Prior revisions specified "The graphite [should] be subjected to an actual factory glazing operation and to such additional test as should be considered necessary by the contracting officer."

6.8 Government limited liability. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

6.9 Changes incorporated. This specification was updated from MIL-DTL-155B to MIL-DTL-155C with Amendment 2 as part of the Munitions Chemical Specification Modernization (MCSM) program. All test procedures were verified and updated as needed to produce valid results. Reference documents have been verified, updated, and hyperlinked to sources. Packaging instructions have been changed. Sampling procedures have been updated. Details of these changes are documented in the ECPs and ERR listed below.

Specification ID	ECP / ERR
MIL-G-155	-
MIL-G-155A	-
MIL-G-155A w/ Amendment 1	-
MIL-G-155A w/ Amendment 2	-
MIL-DTL-155B	Y18Q2002
MIL-DTL-155C	Y21Q2002
MIL-DTL-155C w/ Amendment 1	Y22Q5001
MIL-DTL-155C w/ Amendment 2	Y24Q3007

6.10 Subject term (key word) listing.

Glazing
Lubrication
Pelletization
Munitions Chemical Specification Modernization (MCSM)

6.11 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:19Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army-AR
Navy- OS
Air Force- 170

Preparing activity:
Army- AR
(Project 1376-2025-003)

Review Activities:
Army- AV, MI
Navy- AS
Air Force- 70

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T05:19Z
Check the source to verify that this is the current version before use.